CLOUD CAPSTONE: Design and Evaluate an AWS Solution Using WAF and CAF

Task 1: Review the Existing Architecture

1. Identified Components

- **Frontend Tier**: These are the web application servers that handle all user interactions and process front-facing requests.
- **Backend Tier:** This is the primary database server responsible for securely storing and managing all critical application data.

2. Identified Risks and Weaknesses

- **Single-AZ Deployment (Risk of Total Outage):** Because the workload is confined to a single physical location, any hardware failure or local power outage would cause the entire application to go offline immediately.
- **Open Security Groups (Risk of Unrestricted Access):** Overly permissive firewall rules allow unnecessary traffic between tiers, meaning a security breach in the frontend could lead to unauthorized access to the sensitive backend database.
- **No Backup Strategy (Risk of Permanent Data Loss):** Without an automated system to regularly save data snapshots, any accidental deletion or corruption would likely result in permanent and unrecoverable loss of business information.

Task 2: Evaluate the Workload Using the WAF

1. Operational Excellence

- **Strength:** The organization has established, well-understood manual procedures for managing the application in its current state.
- **Area for Improvement:** Transition from manual deployment to **Infrastructure as Code (IaC)** to ensure environments are consistent and repeatable.
- **Recommendation:** Use **AWS CloudFormation** to automate the provisioning of the entire stack.

2. Security

- **Strength:** Basic perimeter security is currently in place via on-premises firewalls.
- **Area for Improvement:** Implement the **Principle of Least Privilege** by restricting internal traffic so only the frontend tier can communicate with the database.
- **Recommendation:** Utilize **Amazon VPC Security Groups** to act as stateful firewalls for each instance.

3. Reliability

- **Strength:** The application has a proven track record of stability within the current on-premises hardware limits.
- **Area for Improvement:** Eliminate the **Single Point of Failure (SPOF)** caused by the single-AZ deployment.
- **Recommendation:** Deploy the database using **Amazon RDS Multi-AZ** to provide automatic failover across different physical locations.

4. Performance Efficiency

- **Strength:** Current hardware is likely provisioned to handle the application's average historical peak loads.
- **Area for Improvement:** Move away from static hardware sizing to a model that can **dynamically scale** based on real-time demand.
- **Recommendation:** Implement **Amazon EC2 Auto Scaling** to automatically add or remove instances based on traffic.

5. Cost Optimization

- **Strength:** The organization has already paid the capital expenditure (CapEx) for the existing on-premises servers.
- **Area for Improvement:** Switch to a **Consumption-Based Model** to avoid paying for idle resources during periods of low traffic.

- **Recommendation:** Use **AWS Compute Optimizer** to analyze workload patterns and recommend the most cost-effective instance types

Well-Architected Framework (WAF) Evaluation Summary

Pillar	Observation	Improvement Recommendation	Supporting AWS Service
Operational Excellence	Current processes are likely manual and lack visibility.	Implement "Infrastructure as Code" (IaC) to automate environment setup.	**AWS CloudFormation**
Security	Open security groups allow unrestricted access.	Enforce the Principle of Least Privilege by restricting traffic to specific ports/IPs.	**VPC Security Groups**
Reliability	Single-AZ deployment makes the app vulnerable to outages.	Deploy across multiple Availability Zones to ensure high availability.	**Amazon RDS Multi-AZ**
Performance Efficiency	On-premises hardware is often static and cannot handle sudden traffic spikes.	Use dynamic scaling to automatically adjust capacity based on demand.	**Amazon EC2 Auto Scaling**
Cost Optimization	On-premises setups often lead to over-provisioning and wasted capital.	Implement right-sizing and use managed services to reduce administrative overhead.	**AWS Compute Optimizer**

Task 3: Cloud Adoption Framework (CAF) Readiness Summary

1. Business Perspective: Strategy and Outcomes

The Business Perspective evaluates whether the organization's leadership is ready to align cloud technology with actual business growth. To be considered "ready," the company must shift its mindset; instead of seeing the IT department as just a recurring monthly expense, they must view the cloud as a tool to drive value and reach customers faster. A primary enabler for this readiness is Strategy Management, which involves documenting how the move to AWS will support long-term goals like entering new markets or reducing the cost of old hardware. Another essential enabler is Portfolio Management, which helps the company look at its existing on-premises servers and decide which ones should be moved, retired, or rebuilt. Readiness is achieved when the company has a solid business case that proves the migration will provide more than just cost savings. It should show that the move will make the business more "agile," meaning it can respond to market changes in days rather than months. Without this alignment, the migration risks becoming a technical project without a clear business purpose.

2. People Perspective: Culture and Change

The People Perspective is about ensuring the staff is prepared for the shift in how they work. Transitioning to the cloud is a major cultural change, and readiness involves assessing whether the current team has the necessary support to adapt. A key enabler is Cloud Fluency, which simply means building the team's technical confidence through training and hands-on workshops so they aren't intimidated by the new systems. To be "ready," the organization should also focus on Culture Evolution, encouraging a environment where teams can work more independently without waiting for layers of manual approval. The establishment of a Cloud Center of Excellence (CCoE) serves as a major readiness indicator; this is a small group of leaders who help guide the rest of the company through the transition. By using a Change Acceleration approach, the organization can identify potential resistance early and retrain physical server technicians to become cloud engineers. Readiness here means the people are just as prepared for the move as the technology is, ensuring a smooth transition with minimal turnover or confusion.

3. Governance Perspective: Control and Oversight

The Governance Perspective focuses on whether the organization has the right "rules of the road" to manage the cloud effectively. Cloud readiness in this area means moving away from slow, manual check-lists and toward automated oversight. A critical enabler is Cloud Financial

Management, which ensures the company is ready to handle a "pay-as-you-go" billing model. This involves setting up budgets and using tags to track exactly which department is spending money in AWS. Another enabler is Risk Management, where the company identifies potential move-related dangers, like data leaks or service downtime, and plans how to lower those risks using cloud tools. To be fully ready, the organization should implement Application Portfolio Management to ensure the frontend and backend tiers meet the company's internal standards. By putting these guardrails in place before the migration starts, the organization ensures it can stay secure and on-budget without slowing down the developers. Readiness is proven when the company can demonstrate that it has total visibility and control over its new cloud environment.

4. Platform Perspective: Infrastructure and Applications

The Platform Perspective focuses on the technical foundation and whether the organization is prepared to build a modern, scalable home for its application. Readiness involves moving away from fixed physical hardware and instead embracing a platform architecture that can grow as the business grows. A primary enabler for this is Infrastructure as Code, which allows the technical team to set up the network, servers, and databases using a digital template rather than manual clicks. This ensures the environment is well-architected and consistent every time it is deployed, significantly reducing the risk of human error during the move. Another key enabler is Modern Application Development, which encourages the team to use tools such as containers to make the application easier to move and scale across different cloud environments. To be fully prepared, the organization should also have a Continuous Integration and Continuous Delivery pipeline in place. This is an automated system that checks the application code for errors and deploys it automatically. Readiness in this perspective means the organization has moved past slow, manual setups and has a fit-for-purpose platform that is ready to support high traffic and rapid updates without breaking.

5. Security Perspective: Compliance and Assurance

The Security Perspective assesses the organization’s readiness to protect its data in a cloud environment. Readiness here involves moving toward a Zero Trust model, which means the system never assumes a user is safe just because they are logged in. A major enabler is Identity and Access Management (IAM), which ensures that only the right people have access to the backend database. To be truly ready, the organization must implement Data Protection enablers, such as automatically encrypting sensitive customer information so it cannot be read if stolen. Another vital indicator of readiness is Threat Detection; the organization must have automated tools that scan for "bad actors" or weird behavior 24/7. By creating Incident

Response plans ahead of time, the team ensures they are ready to fix a security issue the moment it happens. Readiness is achieved when the organization can prove that its security controls in the cloud are actually stronger than they were on-premises, meeting all legal and industry safety requirements.

6. Operations Perspective: Health and Availability

The Operations Perspective determines if the company is prepared to keep the application running smoothly on a day-to-day basis once the migration is complete. Readiness in this area requires a shift toward observability, which means the technical team uses monitoring dashboards and automated alerts to see exactly how the application is performing in real-time. A major enabler is incident and problem management, which utilizes automation to detect a server crash and fix it, often before a human even realizes there was an issue. To be fully prepared, the organization must also automate its patch management, ensuring that both the frontend and backend servers are always updated with the latest security software without needing manual intervention. By focusing on availability and continuity, the team ensures the application is spread across multiple physical locations to stay online even during a major disaster. Readiness in this perspective means the team can manage a much larger and more complex system with the same number of people, using automation to handle routine maintenance tasks while the staff focuses on higher-value improvements to the application itself.

Task 4: Improved Architecture Description

The proposed architecture transforms the existing two-tier application into a resilient, cloud-native environment. By leveraging a Virtual Private Cloud (VPC) distributed across two distinct Availability Zones, this design eliminates single points of failure while ensuring robust security and automated scalability. The architecture is divided into clear functional layers, separating public-facing entry points from sensitive internal application and database components.

Architectural Components

1. Networking and High Availability

- **VPC and Subnets:** A dedicated VPC spread across two Availability Zones. Traffic is partitioned into Public subnets (for the entry point) and Private subnets (for the core application and database).
- **Availability Zones:** Using two AZs ensures the application remains online even if an entire AWS data center experiences a local power outage or hardware failure.

2. Application and Database Tiers

- **Traffic Management (ALB):** An Application Load Balancer sits in the public subnets to receive user traffic and intelligently distribute it across the frontend servers.
- **Compute Tier (Auto Scaling):** The frontend consists of EC2 instances managed by an Auto Scaling Group, which automatically replaces failed instances and adjusts capacity based on demand.
- **Database Tier (Multi-AZ RDS):** The backend utilizes Amazon RDS with Multi-AZ deployment. This creates a primary database and a synchronous standby replica in separate zones for instant, automatic failover.

3. Security and Compliance

- **Security Groups:** These act as virtual firewalls. The backend database is restricted to only accept traffic from the frontend servers, preventing direct internet access.
- **AWS WAF:** A Web Application Firewall is integrated with the Load Balancer to protect the application from common web exploits like SQL injection.

Project Reflection

Completing this capstone project has provided me with a comprehensive understanding of the strategic shift required when moving from legacy on-premises systems to a well-architected cloud environment. By applying the AWS Well-Architected Framework, I learned that high availability and security are not merely technical configurations but are foundational design principles. Specifically, identifying risks like single points of failure and open security groups taught me the vital importance of data isolation and automated redundancy.

Furthermore, the AWS Cloud Adoption Framework was instrumental in showing that a successful migration depends as much on organizational readiness, such as Cloud Fluency and Governance as it does on the technology itself. I now feel confident in my ability to recommend specific AWS services, like Multi-AZ RDS and Auto Scaling, to solve real-world business challenges. This experience has bridged the gap between theoretical cloud concepts and practical architectural design, ensuring that migrations are secure, scalable, and strategically aligned.